Detector Fine-Tuning for Multi-Object Segmentation and Tracking

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Abstract

We will compare YOLO11s-seg with ByteTrack and detector-assisted SAM 2.1 on a fixed YouTube-VIS 2021 subset. After pretrained baselines, prioritized, budgeted YOLO fine-tuning will test whether improved detections benefit downstream segmentation and identity consistency. Both pipelines share the adapted detector; SAM remains frozen. We evaluate quality and computational cost without assuming improvement.

1 Introduction and research questions

A teaching-video misclassification motivates investigation, not a quantitative claim. We ask: (RQ1) how do tracking-by-detection and detector-assisted mask propagation differ in accuracy and cost? (RQ2) does limited target-domain YOLO adaptation improve detection/segmentation and consequently tracking in both pipelines? Better observations need not solve identity recovery after occlusion. Our contribution is an empirical study, not a new architecture or semantic re-identification system.

2 Related work

YOLO11 predicts masks, boxes, and classes and supports fine-tuning (Ultralytics, 2026). ByteTrack associates detections; it is not a neural model to fine-tune (Zhang et al., 2021). SAM 2.1 propagates prompted masks using video memory, requiring external discovery and labels (Meta AI, 2026). HOTA and IDF1 measure tracking quality (Luiten et al., 2020; Ristani et al., 2016).

3 Methods and adaptation study

Pipeline A uses YOLO11s-seg masks and ByteTrack associations every frame. We will run ByteTrack separately per category with video-wide unique IDs, documenting this class-gated variant. Pipeline B uses the same YOLO checkpoint for periodic box prompts and labels, with frozen SAM 2.1 Hiera Tiny for mask propagation. It receives neither ByteTrack IDs nor ground-truth prompts.

Group	YOLO weights	Downstream pipeline
A0	Pretrained	ByteTrack + YOLO masks
B0	Pretrained	Frozen SAM 2.1
A1	Fine-tuned	ByteTrack + YOLO masks
B1	Same as A1	Frozen SAM 2.1

For B, periodic detections match same-category propagated-mask boxes one-to-one by box IoU. Accepted unmatched detections start tracks; matches refresh prompts. Empty masks yield no output

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box and mark tracks lost. Reappearing propagated masks can recover a retained ID; otherwise unmatched detections receive new IDs. Lost-track expiry, duplicate suppression, and refresh rules will be fixed on validation data. No explicit long-term re-identification is added.

After A0/B0, we will fine-tune one YOLO11s-seg with supervised segmentation losses for A1/B1. A uses its masks and boxes; B uses boxes and labels, so benefits may differ. A pilot fixes the budget: initially at most 30 epochs, 640-pixel input, and memory-appropriate batch size. Validation mask AP50:95 selects the checkpoint. We record sampling, augmentation, optimizer, seed, and head/category-space changes, which limit weight-only causal claims. All runs share the target-category mapping. Detector thresholds and downstream rules stay fixed within A0/A1 and B0/B1; validation tuning precedes evaluation.

4 Dataset and controls

We target 200 labeled YouTube-VIS 2021 training videos covering person, dog, cat, horse, and car (Yang et al., 2021). Seed 550 defines whole-video splits of 120 training, 40 validation, and 40 held-out evaluation videos. Split first, then sample temporally spaced training frames with all target instances retained. Related clips from an identifiable source stay together. Files, coverage, frame lists, mappings, and any reduced counts will be verified and frozen before tuning. These are planned internal splits, not official challenge test sets; upstream pretraining overlap may be unknown.

All conditions process identical ordered annotated frames, using a common external resize and documented model-specific preprocessing. Masks return to annotation coordinates. Training-label conversion will be checked against original masks; evaluation uses original annotations, including late arrivals and disappearances. Held-out data and teaching clips are excluded from training, checkpoint selection, and tuning. Annotation-based failure-case selection will precede inspection of model results.

5 Evaluation and deliverables

Primary evaluation uses mask HOTA with its standard threshold sweep, reporting DetA and AssA, plus mask IDF1 at IoU 0.5. A shared class-aware TrackEval adapter will also report CLEAR false positives, false negatives, and ID switches at mask IoU 0.5 (Luiten and Hoffhues, 2020). Per-category scores combine sequences using each metric's official aggregation, then categories receive equal weight. Per-video results support diagnosis. Synthetic perfect, missed, and switched-ID cases will validate scoring.

Standalone YOLO box/mask AP50:95 and classification-error inspection will distinguish detector changes from tracking changes. We will compare A1-A0 and B1-B0, including regressions and ambiguous outcomes, and inspect occlusion, crossings, and re-entry cases. These are internal measurements, not official leaderboard scores or evidence from one teaching clip. A versus B compares complete systems, including different detector call frequencies, not isolated backbones.

On the same GPU and inputs, we report peak memory, detector calls, and median FPS from three synchronized, warmed-up runs. Timing covers decoding through masks/IDs, excluding model loading, rendering, and scoring. Training GPU-hours are separate. Deliverables include configurations, manifests, complete predictions, training logs, four-condition results when feasible, synchronized videos, and a report.

6 Feasibility and milestones

Short inference works on our RTX 3080 (10 GB); full SAM tracking and YOLO training need pilots. Rutgers iLab use follows access/scheduling rules; availability is not assumed. With data/environment ready, estimated milestones are baseline/data/evaluator work (3-5 working days), adaptation pilot and experiments (2-4), then reporting (1-2), revised using measured costs. Both pretrained baselines on the entire evaluation split are the minimum deliverable; fine-tuning is prioritized, with resource-driven omissions documented. SAM/DeAOT training and additional pipelines are out of scope. Success does not require gains.

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